Industrial Revolution

04 Sep 05:30
Duty six : Purple

<table>
<tr><th>Start</th><th>Time</th><th>Activity</th><th>Requirements</th><th>Instructions</th><th>Scouter</th></tr>
<tr><td>04 Sep 05:30</td><td>5</td><td>Activities : Opening</td><td>Register, beans, flag, totem and skin</td><td>Grand Howl
Flag Break
Register
Inspection - belts and shoes</td><td>Akela</td></tr>
<tr><td>04 Sep 05:35</td><td>10</td><td>Game : Human Machines</td><td></td><td>Six corners represent the six simple machines. Call out a machine and Cubs race to the correct corner, then perform an action: lever = pretend to lift a boulder; pulley = pull rope; wheel & axle = spin; wedge = chop motion; inclined plane = climb; screw = turn downwards. After a few rounds call out everyday objects instead: ramp, knife, flagpole, screwdriver, seesaw, bicycle wheel. Cubs decide which machine it uses. Instead of six directions, the Cubs can just stop and do the actions.</td><td>Akela</td></tr>
<tr><td>04 Sep 05:45</td><td>10</td><td>Activities : Machine Detective</td><td>None</td><td>Cubs to identify everyday objects that are simple machines and tell us one thing needed to keep it working. They can look around the hall but Scout leader can also provide some items for them to try and guess what simple machine they are.

For example:

<table><tr><th>Object</th><th>Machine</th><th>Maintenance</th></tr><tr><td>Bicycle</td><td>Wheel & axle</td><td>Keep tyres inflated; lubricate moving parts</td></tr><tr><td>Scissors</td><td>Lever/wedge</td><td>Keep clean and dry; sharpen when necessary</td></tr><tr><td>Door hinge/handle</td><td>Wheel & axle / lever</td><td>Tighten screws; lubricate if stiff</td></tr><tr><td>Screwdriver & screws</td><td>Wheel & axle / screw</td><td>Keep dry and prevent rust</td></tr><tr><td>Pulley clothesline</td><td>Pulley</td><td>Check rope and pulley for wear</td></tr><tr><td>Garden shears</td><td>Lever/wedge</td><td>Clean, oil pivot and sharpen blades</td></tr></table></td><td>Akela</td></tr>
<tr><td>04 Sep 05:55</td><td>20</td><td>Activities : Catapult</td><td>Popsicle sticks (or craft sticks) (9 each)
Rubber bands (5 each)
Plastic Spoon (1 each)
Styrofoam Balls (or Pom Poms or marshmallows or anything that fits in the spoon)</td><td>1. Make a stack of 7 popsicle sticks and use rubber bands to tie them together on both ends.

2. Make a stack of 2 popsicle sticks and use a rubber band to tie them together on one end only.

3. Pull the 2 popsicle sticks apart and wedge the stack of 7 popsicle sticks between them.</td><td>Akela</td></tr>
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				4. Use two rubber bands, secure the plastic soon to the upper popsicle stick. 5. Place the ball onto the spoon. 6. Hold the catapult with one hand, use the other hand to push down the spoon. 7. Release the spoon and watch the ball fly!	
04 Sep 06:15	5	**Activities :** Juice and biscuits		Juice and biscuit break	Akela
04 Sep 06:20	25	**Activities :** Machine Challenge	Plank/box Toy cars Bucket Rope Stones Nuts and Bolts	Cubs to complete the various challenges. They can divide into smaller groups if needed and do each challenge separately. Lever – Lift the Load Place a heavy box or bucket on the floor. Give Cubs a broomstick and a wooden block as the fulcrum. They have to lift the load without picking it up. Move the fulcrum closer to and farther from the load so they discover that its position changes how much effort is needed. Key idea: A lever lets us move a heavy load using less effort. Examples: seesaw, bottle opener, crowbar, scissors. Pulley – Rescue the Supplies Run rope through a pulley or over a pole attached to something secure or even a tree branch. Cubs lift a bucket containing beanbags or another small weight. If possible, let them first lift the bucket directly and then with the pulley. Key idea: A pulley changes the direction of the force and can make lifting easier. Inclined Plane – Mountain Supply Run Place one end of a plank on a chair or sturdy box. Cubs race cars down the ramp. Change the angle to see what effect it has (cars go faster/further if incline is higher). Key idea: A ramp lets us use less force over a longer distance. Examples: wheelchair ramp, stairs, loading ramp, slide. Screw – Up We Go Give each Cub a large bolt and nut. Race to see who can move the nut from one end to the other. Then show that the thread is essentially an inclined plane wrapped around a cylinder. If you have wood and appropriate supervision, Cubs can also drive a screw into soft wood with a screwdriver. The nut and bolts can be done as a relay race.	Akela

				Key idea: A screw changes turning movement into movement forwards or backwards. Examples: screws, jar lids, bolts, light bulbs	
04 Sep 06:45	5	**Activities :** Closing	Totem, Skin Badges, certificates	Announcements Badge handouts Grand Howl Flag Down Prayer Dismiss	Akela
04 Sep 06:50	10	**Activities :** Industrial Revolution inventions	Meccano models of various machines (traction engine, mill, etc.)	Show the Cubs the different more complex machines that formed the basis of the first industrial revolution. Even if they were more complex, they included some of the simple machines that we learnt about. Ask Cubs to identify some of them incorporated into these engineering wonders. Just like the inventors of the industrial revolution, we want the Cubs to invent something. But, because they are Cubs, it should be something fun. Each Cub to design a game for the other Cubs that incorporates at least one of the simple machines. They can use their imaginations and combine things as well - they don't actually have to make it or play it right now - they can go home and bring back their ideas to earn their badge.	Akela

Programme prepared by Sharon Venn of 1st Randburg on 03 Sep 21:00